Elton Chang

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EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA

University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

EXPERIENCE

Co-Founder 2026 - Present
AMOS
- Co-developed a Rust/Axum control-layer prototype for AI data analysis with parsed SQL validation, role-based tool permissions, independent statistical execution, and evidence-linked claim replay.
- Designed versioned REST/OpenAPI APIs with bearer authentication and idempotent SQLite transactions; packaged a non-root Docker image with hardened Compose and configured GitHub Actions test/benchmark gates.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA
- Reduced configuration drift across 15+ non-production hosts by provisioning test environments with Terraform.
- Cut anomaly detection time by 40% with telemetry, alerting, and dashboards for backup and replication pipelines.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA
- Built a Java/SQL file management system with scheduled Python jobs for file ingestion, metadata indexing, and lifecycle tracking.
- Improved onboarding and adoption by 35% with architecture and workflow documentation for developers and users.

PROJECTS

Assay | Sandboxed Analytics and Atomic Release Control
- Built a Python/DuckDB research prototype for synthetic tasks with SQL AST allowlists, network-disabled Docker workers, resource limits, and content-addressed evidence.
- Implemented revision-bound approvals and atomic SQLite source-epoch checks to block stale report releases.

Squid | C++ Firmware and ROS 2 Sensor Pipeline Sep 2026
- Implemented C++17 RP2040 firmware for SPI encoder and multiplexed strain acquisition, with binary USB framing, parity checks, angle unwrapping, and host-side protocol tests.
- Built Python ROS 2/MCAP recording with causal timestamp gating and jitter diagnostics; added parser resynchronization, emulated serial stalls, and fault-injection tests.

Torflix | Asynchronous BitTorrent Engine and Hash-Verified Streaming
- Implemented a Python asyncio BitTorrent engine with rarest-first/sequential piece selection, bounded request pipelines, SHA-1 verification, and hash-revalidated resume state.
- Mapped torrent offsets across files under async locks; served contiguous verified byte ranges through FastAPI, with WebSocket progress and a React/TypeScript playback client.

Zoe | Full-Stack Mobile Platform with Indexed Search and Federated Identity
- Built a TypeScript/Hono API and Expo/React Native client with PostgreSQL/Prisma, GIN-indexed full-text search, Argon2id passwords, and hashed opaque sessions.
- Implemented S3-compatible presigned uploads with MIME/size checks and AES-256-GCM OAuth token storage.

OPEN SOURCE CONTRIBUTIONS

- Merged contributions include vLLM TPU Inference (KV-cache connector), Apache DataFusion (Rust SQL engine), SQLFluff (SparkSQL parser), Career Ops (JavaScript), and HoloViz Panel (three UI-state fixes).
- Opened inference-infrastructure pull requests to Dynamo (VRAM-aware pytest workers), ONNX Runtime (CUDA CMake), and vLLM Production Stack (Kubernetes probe).

TECHNICAL SKILLS

Languages & Web: Python, TypeScript, JavaScript, Rust, Java, Go, C/C++, SQL, React, Next.js, Node.js, FastAPI
AI, Data & Cloud: LLM APIs, RAG, MCP, PyTorch, PostgreSQL, MongoDB, Docker, Kubernetes, Terraform, GCP